1.					
	a	b	c	abc	One correct set a, b & c B1
	2	**15**	**3**	**90**	
	3	**9**	**8**	**216**	Two correct rows M1
	4	**1**	**15**	**60**	
	All values of abc are multiples of 6				Fully correct + statement A1
					(3 marks)

2.(a)	E.g. $64\times\left(-\frac{1}{2}\right)^3=-8$ ✓ *	M1, A1* **(2)**
(b)	Finds the value of a. E.g. $64\div\left(-\frac{1}{2}\right)^2=256$	M1, A1
	Uses $S_\infty=\frac{a}{1-r}=\frac{256}{1-\left(-\frac{1}{2}\right)}=\frac{512}{3}$	M1, A1 **(4)** **(6 marks)**

	$f(x)=\left(3x^2-4x-5\right)\left(x-k\right)-5$	
3 (a)	States -5	B1 **(1)**
(b)	Sets $f(-2)=25\rightarrow\ \left(3\times4-4\times-2-5\right)\left(-2-k\right)-5=25$	M1
	$15\left(-2-k\right)=30\Rightarrow-2-k=2\Rightarrow k=-4$ *	A1* **(2)**
(c)	$\left(3x^2-4x-5\right)\left(x+4\right)-5=3x^3+8x^2-21x-25$	B1
	Attempts $3x-1\overline{)3x^3+8x^2-21x-25}$ to achieve quotient of $...x^2+...x+...$ and a remainder that is a constant	M1
	Quotient = x^2+3x-6 OR Remainder = -31	A1ft
	Quotient = x^2+3x-6 AND Remainder = -31	A1 **(4)** **(7 marks)**

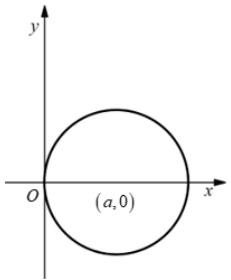
4. (a)	$\log_3(a+1)-\log_3 a=4 \Rightarrow \log_3\left(\frac{a+1}{a}\right)=4$	M1
	$\Rightarrow \left(\frac{a+1}{a}\right)=3^4$	A1
	$\Rightarrow a+1=81a \Rightarrow a=\frac{1}{80}$	dM1, A1
		(4)
(b)	$\text{Area} \approx \frac{1}{2}\left\{\log_3\left(\frac{2}{1}\right)+\log_3\left(\frac{6}{5}\right)+2\left(\log_3\left(\frac{3}{2}\right)+\log_3\left(\frac{4}{3}\right)+\log_3\left(\frac{5}{4}\right)\right)\right\}$	M1, A1
	$=\frac{1}{2}\log_3\left(\frac{2}{1}\times\frac{6}{5}\times\frac{3^2}{2^2}\times\frac{4^2}{3^2}\times\frac{5^2}{4^2}\right)=\frac{1}{2}\log_3 15=\log_3\sqrt{15}$	dM1, A1
		(4)
(c)	States 'increase the number of strips'	B1
		(1)
		(9 marks)

5. (a)	$u_2 = 1-\frac{1}{4}=\frac{3}{4}, \quad u_3 = 1-\frac{4}{3}=-\frac{1}{3},$	M1 A1
	$u_4 = 1-\left(-\frac{3}{1}\right)=4$ with statement saying the sequence repeats every 3 terms o.e	A1*
		(3)
(b)	$\sum_{n=1}^{180}(5n+3)=\frac{180}{2}(8+903)=(81990)$	M1
	$\sum_{n=1}^{180}(u_n)=60\times\left(4+\frac{3}{4}+-\frac{1}{3}\right)=(265)$	M1
	$\sum_{n=1}^{180}(5n+3+u_n)=81990+265=82255$	A1, A1
		(4)
		(7 marks)

6 (i) (a)	Uses $\sin^2\theta+\cos^2\theta=1$ with $\cos\theta=\frac{1}{\sqrt{5}}\Rightarrow\sin^2\theta=\frac{4}{5}$	M1
	$\Rightarrow\sin\theta=-\frac{2}{\sqrt{5}}$	A1
(b)	Uses $\tan\theta=\frac{\sin\theta}{\cos\theta}$ with $\cos\theta=\frac{1}{\sqrt{5}}$ and their $\sin\theta=-\frac{2}{\sqrt{5}}$	M1
	$\Rightarrow\tan\theta=-2$	A1
		(4)
(ii) (a)	1 (m)	B1
		(1)
(b)	$30t-40=180\Rightarrow t=7.33$ Hence 7:20 **am**	M1, A1, A1
		(3)
(c)	$4+3\cos(30T-40)^\circ=3.5\Rightarrow\cos(30T-40)^\circ=-\frac{1}{6}$	M1
	$\Rightarrow 30T-40=99.6,\ 260.4,\ \underline{\underline{459.6}}$	A1
	$\Rightarrow T=\frac{"459.6"+40}{30}=16.65$	dM1, A1
		(4)
		(12 marks)

7 (a)	States either $kn=-24$ (1) or $\frac{n(n-1)}{2}k^2=270$ (2)	M1
	States both $kn=-24$ (1) and $\frac{n(n-1)}{2}k^2=270$ (2)	A1
	Substitutes $k=-\frac{24}{n}$ in equation (2) $\Rightarrow \frac{n(n-1)}{2}\left(-\frac{24}{n}\right)^2=270\Rightarrow n=...$	M1
	$n=16$	A1
	Uses their $n=16$ in $kn=-24\Rightarrow k=-\frac{24}{16}=-\frac{3}{2}$	dM1, A1
		(6)
(b)	$p=\frac{n(n-1)(n-2)}{3!}k^3=\frac{16\times15\times14}{6}\times\left(-\frac{3}{2}\right)^3=...$	M1
	$=-1890$	A1
		(2)
		(8 marks)

8. (a)	$-5 < x < \frac{2}{3}$	M1, A1 **(2)**
(b)	$(2x-7)$	B1 **(1)**
(c)	$f'(x) = 2(3x-2)(x+5) = 6x^2 + 26x - 20$	
	$f(x) = 2x^3 + 13x^2 - 20x + c$	M1, A1
	$x = \frac{7}{2}, \quad y / f(x) = 0 \Rightarrow c = (-175)$	dM1, A1
	$f(x) = 2x^3 + 13x^2 - 20x - 175 = (2x-7)(x^2 + 10x + 25)$	ddM1
	$= (2x-7)(x+5)^2$	A1
		(6) **(9 marks)**

9(a)		B1 **(1)**
(b)	$(x \pm a)^2 + y^2 = ...$	M1
	$(x \pm a)^2 + y^2 = a^2$	A1
	Uses $(5, 6)$ in $(x \pm a)^2 + y^2 = a^2$ to form and solve an equation in a	dM1
	E.g. $(5-a)^2 + 36 = a^2 \Rightarrow 10a = 61 \Rightarrow a = 6.1$	
	$(x-6.1)^2 + y^2 = 6.1^2$	A1
		(4) **(5 marks)**

10 (a)	$y = 2x + \frac{64}{x^2} - 3$ $\frac{dy}{dx} = 2 - \frac{128}{x^3}$	M1, A1
	Attempts to solve $\frac{dy}{dx} \Rightarrow x^3 = 64 \Rightarrow x = 4$ *	dM1, A1* **(4)**
(b)	$\int 2x + \frac{64}{x^2} - 3 \, dx = x^2 - \frac{64}{x} - 3x \quad (+c)$	M1, A1
	Finds the y values at both $x = 2$ and $x = 4$. $M = (0, 17)$ and $N = (0, 9)$	B1
	Full attempt at area = $\left[x^2 - \frac{64}{x} - 3x \right]_2^4 - 9 \times 2 + 2 \times (17 - 9) = 22 - 18 + 16 = 20$	dM1, A1 **(5)** **(9 marks)**